

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.4: Electrostatics — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.4: Electrostatics
Driving Question	DRIVING QUESTION: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Mysterious Balloon: What Invisible Force Is at Work?	After 20 strokes on dry hair, the rubbed balloon made hair strands rise and follow it up to about 5 cm, stuck to the wall for about 3 minutes, and made 1 cm x 1 cm paper pieces jump up when held about 2 cm above them, with some falling off again. Two rubbed balloons on threads pushed apart. An unrubbed balloon did nothing. The effect is weak or absent on humid days (Mombasa, about 80%) and strong on dry days (Nairobi, about 35%).	Rubbing makes the balloon able to pull on hair, wall and paper without touching them, and to push away another rubbed balloon. The effect is not the same as a magnet, it fades with time and it depends on the weather. We now have many questions on our DQB.	So far we can only describe what happens: something invisible passes between the rubbed balloon and other objects, and rubbing and dry air seem to matter. Our initial models are hunches, and we do not yet know where the effect comes from or why the balloon sometimes attracts and sometimes repels.
Lesson 2 What Are Electric Charges? Atoms, Protons, Electrons and the Rule of Attraction and Repulsion	Two polythene strips rubbed with wool/fur repel; two perspex rods rubbed with silk repel; a polythene strip and a perspex rod attract; unrubbed strips and rods do nothing. Proton charge is $+1.6 \times 10^{-19}$ C, electron charge is -1.6×10^{-19} C and neutron charge is 0 C.	Atoms contain positive protons and neutral neutrons in the nucleus with negative electrons around it. A neutral atom has equal numbers of protons and electrons, so the total charge is 0 C. Like charges repel and unlike charges attract through the electrostatic force.	The two rubbed balloons in Lesson 1 pushed apart because they carry the same type of charge, while the balloon and hair attracted because they carry opposite types. We still need to find out how rubbing gives objects their charge and why neutral objects such as paper and the wall are also attracted.
Lesson 3 Charging by Friction: Who Wins the Electrons?	We rubbed pairs of materials (glass with silk, ebonite with fur, perspex with wool/fur, polythene with silk, balloon with hair) and tested them against a negative polythene strip and a positive perspex rod. Rod and cloth had opposite signs, shown by	Friction transfers electrons from one material to the other. Protons stay in the nucleus. The sign depends on the material pair, following the triboelectric order: dry human hair (+), glass, perspex, wool/fur, silk, polythene, ebonite, latex balloon (-).	When I rub the balloon on my hair, electrons transfer from the hair to the balloon because latex holds electrons more strongly. The balloon becomes negative and the hair positive. The hair strands, all positive, repel each other and rise, and they

	repulsion. The balloon was negative and the hair positive. Unrubbed objects showed no effect.	Charge is conserved: the balloon gains about 1 nC (6×10^9 electrons) and the hair loses the same amount, out of about 10^{23} atoms.	are attracted to the negative balloon. Charge is not created. Still to explain: why the balloon sticks to the neutral wall and attracts paper, and why humidity makes it fail.
Lesson 4 Conductors and Insulators: Why Does the Balloon Keep Its Charge While a Metal Spoon Does Not?	The bulb lit with copper wire, aluminium foil, a steel spoon and graphite. It did not light with a plastic ruler, dry wood, rubber, a glass rod or dry cotton cloth in the 3 V circuit.	Conductors have free (delocalised) electrons that can move through the material; insulators have tightly bound electrons. Only electrons move in solids. The latex balloon is an insulator, so the electrons it gains from the hair stay on it.	The balloon is negative and holds its extra electrons (about 6×10^9 for 1 nC) because latex is an insulator, so the charge cannot move away and the balloon stays charged and sticks for about 3 minutes on a dry day.
Lesson 5 Charging by Contact: Why Do Paper Pieces Touch the Balloon and Then Fall Off?	The leaf of a neutral electroscope stayed at about 30 degrees after a negative polythene strip touched the cap and was removed (0 degrees before; about 30 degrees also when the strip was only near). Touching the cap with a finger returned the leaf to 0 degrees. Paper pieces jumped up to the rubbed balloon and some fell off again after touching it.	A conductor touched by a charged object gains the SAME sign of charge by transfer of electrons. Free electrons in the conductor let the charge spread to the leaf, and like charges repel, so the leaf stays raised. Charge is transferred, not created.	The negative strip passed extra electrons to the cap, stem and leaf, so they all became negative and repelled each other. In the same way, a paper piece that touches the negative balloon gains electrons, becomes negative, and is repelled, so it falls off. This adds one more step to the balloon-and-hair explanation, though why neutral paper is attracted at first is still to be explained.
Lesson 6 Charging by Induction and Earthing: Charging Without Touching	A neutral electroscope with a negative polythene strip brought near the cap, not touching: the leaf rose to about 30 degrees and fell back to 0 degrees when the strip was removed. With the strip near and a finger touching the cap (earthing), the leaf fell to 0 degrees. After removing the finger and then the strip, the leaf rose to about 30 degrees and stayed up. Neutral paper pieces jumped to the negative balloon and the balloon stuck to the wall, with no repulsion seen.	A charged object can separate charge in a neutral object without touching it (induction). Earthing lets electrons flow to or from the Earth, and charging by induction with earthing gives the OPPOSITE sign, while contact gives the SAME sign. Only electrons move, and charge is not created but transferred.	The negative balloon repels electrons in the neutral paper and wall, leaving the near surface positive and the far side negative. The nearer positive charge is attracted more strongly than the farther negative charge is repelled, so a neutral object is attracted but never repelled. This explains why the paper jumps and the balloon sticks to the wall. On humid days, such as in Mombasa, the balloon's charge leaks away, so the effect is weak or short-lived.
Lesson 7 Distribution of Charge on Conductors: Outer Surfaces, Points, Sparks and Lightning Conductors	We observed that a charged hollow metal can gave an electroscope leaf reading of about 40 degrees when the proof plane touched the outside and 0 degrees when it touched the inside. On the pear-shaped conductor, the pointed end gave about 45 degrees and the flat end about 20 degrees.	We learned that the free electrons on a charged conductor repel each other and move to the outer surface, and that charge is most concentrated at points. At a point, the strong field ionises the air and charge leaks away (point discharge), while earthing lets charge flow safely to the ground.	This explains why a woollen jersey shock in dry Nairobi is a small spark (a discharge) and why lightning conductors, with a pointed rod joined by thick copper strip to a buried earth plate, protect buildings in Kericho and Kisumu. It also adds to our model of how electrons, not created charge, move and build up around the balloon, hair, wall and paper.
Lesson 8 Detecting and Measuring	Record what you observed with the gold-leaf electroscope: the leaf at 0 degrees	Write what you learned: a gold-leaf electroscope detects charge because the	Explain how this helps answer the Driving Question: the rubbed balloon is shown to be

Charge: The Gold-Leaf Electroscope	when neutral; about 30 degrees with a negative polythene strip (20 strokes); the leaf staying up after contact; about 50 degrees with 40 strokes; the angle going above 30 degrees with a negative object and below 30 degrees with a positive object; and 0 degrees after earthing with a finger.	leaf and stem carry the same sign and repel; a bigger leaf divergence shows more charge; an increase on a charged electroscope shows the same sign and a decrease shows the opposite sign; earthing lets electrons flow to or from the Earth and discharges the electroscope.	charged because it makes the leaf diverge, its negative sign can be tested, and the amount of charge relates to the number of strokes because more electrons are transferred, not created.
Lesson 9 Applications and Consolidation: From the Mysterious Balloon to Lightning Conductors, and the Final Explanation	What did we observe? (Examples: the rubbed balloon raises hair and sticks to the wall for about 3 minutes at about 35% humidity but only under 30 seconds at about 80%; paper pieces jump to a rubbed polythene strip; the lightning conductor has a pointed rod and a thick copper strip to an earth plate; a fuel tanker trails an earthing chain; a photocopier and a precipitator use attraction between unlike charges.)	What did we learn? (Examples: charges are not created, electrons are transferred by friction and total charge is conserved; the balloon gains about 1 nC, about 6×10^9 electrons, out of about 10^{23} atoms; unlike charges attract and like charges repel; neutral objects are attracted by induction; charge crowds at points (point discharge) and earthing lets electrons flow to or from the ground; humidity lets charge leak away.)	How can we explain the phenomenon? (Write your Final Explanation: when the balloon is rubbed on dry hair, electrons transfer from hair to latex because latex holds electrons more strongly, so the balloon is negative and the hair positive; the positive hair strands repel each other and rise and are attracted to the balloon; the negative balloon induces opposite charge on the near side of the neutral wall and paper so they attract; on dry days charge leaks slowly but on humid days water vapour lets it escape; technology such as lightning conductors, tanker earthing, photocopiers and precipitators uses the same ideas.)

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